

The Development Application Base of Engineering Talents (iDABET)
Project of the Industrial Development Administration, Ministry of Economic Affairs
**2025 Practical Ability Development Implementation Unit Project –
Engineering Talent Practical Ability Development Record Form**

Project No. of the Practical Ability Development Implementation Unit: DABET11403

No. of Engineering Talent: 40

Name of the Practical Ability Development Implementation Unit: 財團法人金屬工業研究發展中心/光電技術組

Project Name of the Practical Ability Development Implementation Unit: 半導體產業之真空設備與製程能力發展與人工智慧控制應用

Name of the Engineering Talent: 蘇翼

Item	Date	Instructor	No. of Hours (Note 1)	Type of Practical Ability Development Topic (Note 2)	Topic	Description
1	7/1	李宗信	7	1	Indoor Positioning System Research (Thesis)	Independent Study: Replaced PyQt5 with pure Tkinter for stability and unified architecture.
2	7/2	李宗信	7	1	Indoor Positioning System Research (Thesis)	Independent Study: Refactored to update plot data only, greatly improving performance and responsiveness.
3	7/4	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Project briefing
4	7/7	李宗信	7	1	Indoor Positioning System Research (Thesis)	Independent Study: Implemented accurate MODBUS function codes and data structures for reliable hardware communication.
5	7/8	李宗信	7	1	Indoor Positioning System Research (Thesis)	Independent Study: Separated tag and anchor data into dedicated managers for data consistency.
6	7/9	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Used root.after() to enforce correct GUI initialization before launching background tasks.
7	7/10	李宗信	2	1	From Linguist to Auto-Dealer	Explaining how life can be turned around for the better.
8	7/10	李宗信	2	1	Presentation rehearsal	Rehearsal of End of Internship Presentation
9	7/11	李宗信	7	1	Indoor Positioning System Research (Thesis)	Independent Study: Centralized all visualization logic into a reusable, self-contained class.

10	7/15	李宗信	7	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development-> Redesigned bottom cover to fit Li-ion battery and maintain ground clearance.
11	7/16	李宗信	2	1	Indoor Positioning System Research (Thesis)	Independent Study: Project briefing
12	7/17	李宗信	2	1	Taguchi's method for Quality Engineering	I learned how to work smart using Taguchi methods to get good results in quality control.
13	7/17	李宗信	2	1	Presentation rehearsal	Rehearsal of End of Internship Presentation
14	7/18	李宗信	6	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development-> Added 12V-5V buck converter for stable Raspberry Pi 4 power.
15	7/21	李宗信	6	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development->Top cover redesigned to house Pi, PCA9685s, and cable routing.
16	7/22	李宗信	5	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development-> Integrated GY-85 IMU for pitch, roll, and yaw sensing.
17	7/23	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development-> Mounted dual PCA9685s with standoffs for isolated servo control.
18	7/24	李宗信	8	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development-> Damped IMU mount to reduce vibration and improve sensor accuracy.
19	7/25	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development->Extended wiring harness with labeled, color-coded servo cables.
20	7/28	李宗信	3	1	Indoor Positioning System Research (Thesis)	Independent Study: Hexapod development->Separated power lines for logic and motors to improve stability.
Total Hours on Practical Ability Development Topics			Total ____96__ hours.			

Please add rows to the table as needed.

Notes: (1) The hours for implementing practical ability development topics should be filled in according to the actual implementation status between March 1 and August 31.

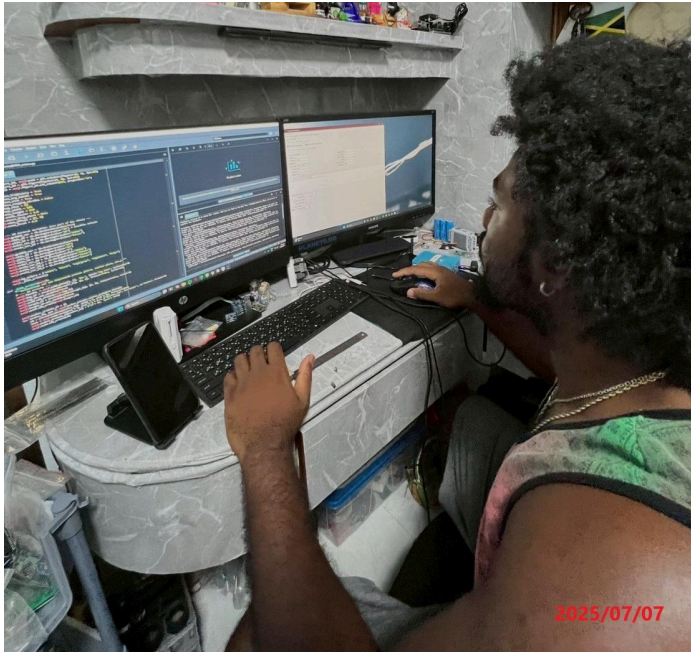
(2) Type of practical ability development topics: 1. Practice-based; 2. Industry collaboration-based.

(3) Please attach at least 6 dated photos (yyyy/mm/dd) as evidence of the engineering talent's participation in practical ability

development topics or activities.

- (4) Both the instructor and engineering talent shall sign at the bottom of the record form.
- (5) To implement the “2025 Development Application Base of Engineering Talents (iDABET) Project”, the Industrial Development Administration, Ministry of Economic Affairs will collect, process and use the personal data of the party involved without notifying the said party, as prescribed in Subparagraph 2 of Paragraph 2 of Article 8 of the Personal Data Protection Act.

【Photos of Participation in Practical Ability Development Topics or Activities】



Code Development

Currently debugging the GUI. The 2D and 3D positioning monitor isn't functioning as expected due to improper handling of the Modbus protocol. I'm working on modifying the algorithm to ensure proper data processing and restore full functionality.



Project Briefing

Today, we had our weekly project update along with a rehearsal in preparation for the main rehearsal on the 17th. My advisor reviewed some of the content and pointed out areas that need to be revised.



Taguchi's Method on Quality Engineering

We learned about the Taguchi Method in Quality Engineering, a statistical approach used to improve product quality by minimizing variation and identifying optimal conditions through robust design. It focuses on designing experiments (DOE) to systematically test different variables and reduce the effects of noise factors, ultimately enhancing performance and consistency in manufacturing and quality control.



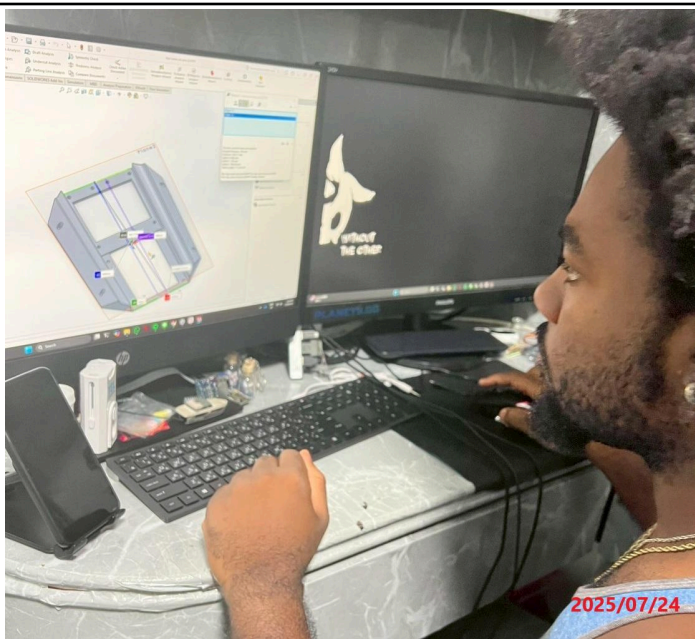
Rehearsal for the Final Presentation

Today was my rehearsal presentation. I was quite nervous, as I'm still building confidence in presenting in Chinese. That said, it went fairly well overall. I've realized I need to work on improving my timing and pacing when explaining the content to ensure clarity and flow.



Hexapod Development (Software based)

Currently refining the software algorithm for my hexapod robot to ensure smoother synchronization with the GUI. The goal is to enhance real-time communication between the robot's movement controls and the visual interface, allowing for more accurate feedback, responsive command execution, and better user interaction. This includes debugging data flow, optimizing update intervals, and improving how sensor data and servo angles are displayed within the GUI.



Hexapod Development (3D design)

Currently designing an improved top cover in SolidWorks to securely house the Raspberry Pi 4B and PCA boards for the prototype robot. I'm ensuring all dimensions and mounting points are precise to guarantee a clean and functional fit before moving to 3D printing. As always... measure twice, print once.

Notes:

1. Please add rows to the table as needed. Each photo shall be dated at the bottom right, and each photo shall be accompanied by descriptions below it to explain the topic.
2. Please provide the original files of 1 to 3 photos (JPEG files, each above 1MB in size and with a resolution above 300 DPI).